UNIVERZITET U NOVOM SADU
FAKULTET TEHNIČKIH NAUKA
KATEDRA ZA AUTOMATIKU I UPRAVLJANJE SISTEMIMA

Modeli jednostavnih električnih i elektro-mehaničkih Sistema

Modeli fizičkih sistema

Modeliranje i simulacija sistema

Upravljanje, modelovanje i simulacija sistema

Električni sistemi

Veličine i elementi

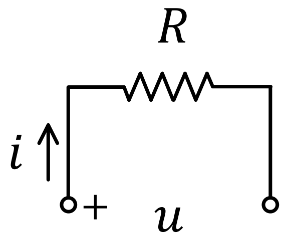
Osnovne fizičke veličine:

- napon u $[V]$ i
- jačina električne struje i $[A]$.

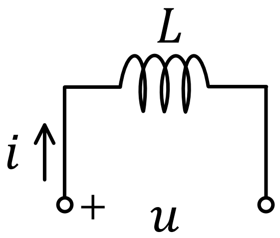
- električna snaga $p = ui$ $[W]$ i
- električna energija $E = \int p(t)dt$

Pasivni elementi:

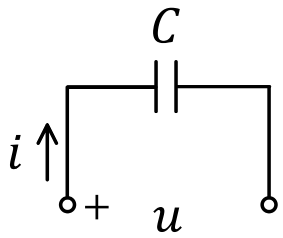
- otpornik otpornosti R $[\Omega]$ $\quad u(t) = Ri(t)$
- kalem induktivnosti L $[H]$ $\quad u(t) = L\frac{di_L(t)}{dt}$
- kondenzator kapaciteta C $[F]$ $\quad i(t) = C\frac{du_C(t)}{dt}$



Otpornik



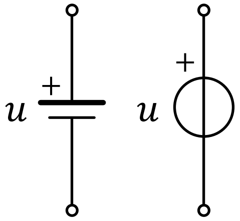
Kalem



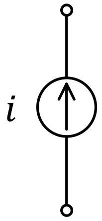
Kondenzator

Aktivni elementi:

- Idealan naponski izvor
- Idealan strujni izvor



Idealni
naponski izvori



Idealan
strujni izvor

Zakonitosti

- Omov zakon

$$i = \frac{u}{R}$$

- I Kirhofov zakon: algebarska suma struja koje utiču u ma koji čvor kola jednaka je nuli

$$\sum_{po\ granama\ k\ u\ čvoru} i_k = 0$$

- II Kirhofov zakon: algebarska suma padova napona u bilo kojoj petlji električnog kola ja jednaka nuli

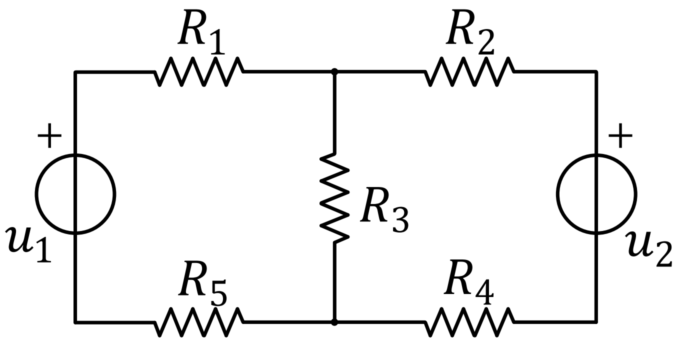
$$\sum_{po\ elementima\ k\ u\ konturi} u_k = 0$$

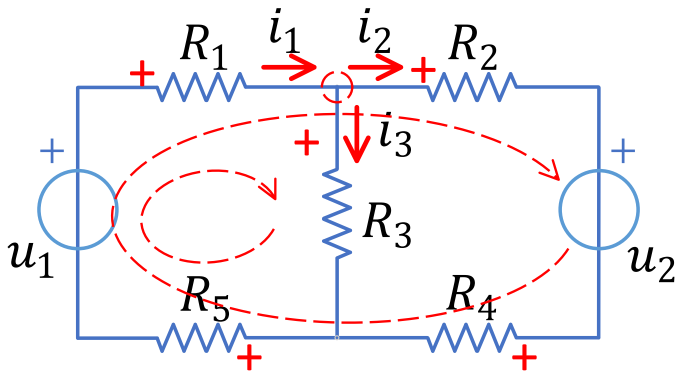
- Alternative tehnike formiranja modela
  - Metoda potencijala čvorova
  - Metoda konturnih struja

  upotrebljavaju se često jer se dobija manji broj jednačina u odnosu na direktnu upotrebu Kirhofovih zakona

Primer 1 – Električno kolo sa otpornicima

Zadatak:
Napisati model za el.kolo sa slike.





Formiranje modela primenom Kirhofovih zakona:

$$i_1 - i_2 - i_3 = 0$$

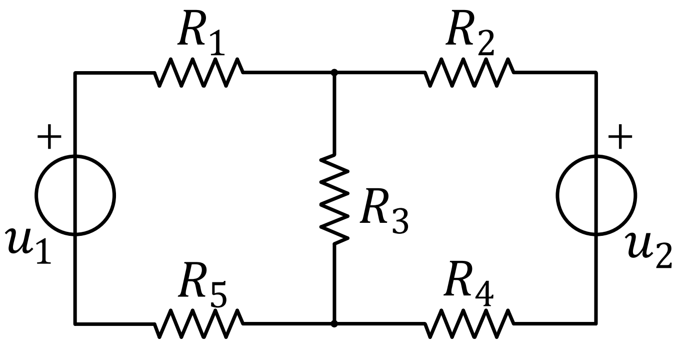
$$u_1 - (R_1 + R_5)i_1 - R_3 i_3 = 0$$

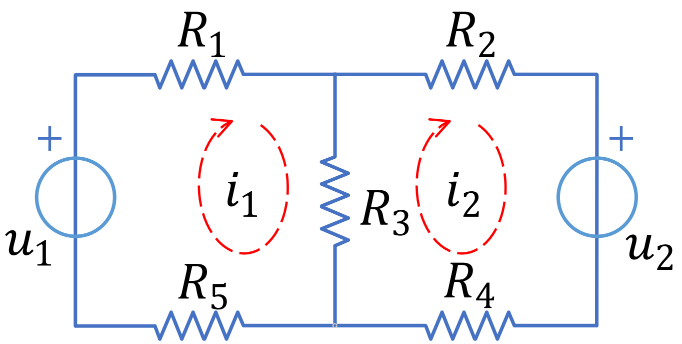
$$u_1 - (R_1 + R_5)i_1 - (R_2 + R_4)i_2 - u_2 = 0$$

U matričnom zapisu:

$$\begin{bmatrix} 1 & -1 & -1 \\ -R_1 - R_5 & 0 & -R_3 \\ -R_1 - R_5 & -R_2 - R_4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ u_1 \\ u_1 - u_2 \end{bmatrix}$$

Primer 1 – nastavak ...





Konturne struje:

$$R_1 i_1 + R_3(i_1 - i_2) + R_5 i_1 = u_1$$

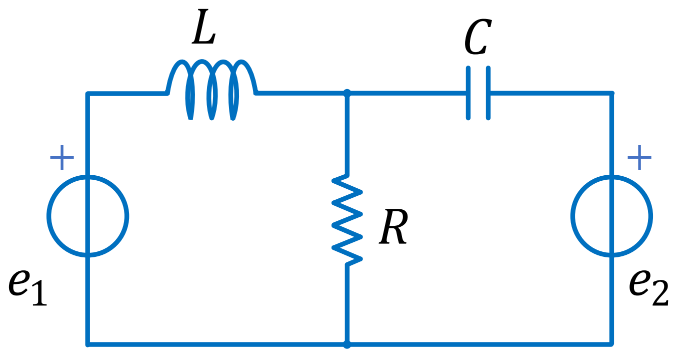
$$R_4 i_2 + R_3(i_2 - i_1) + R_2 i_2 = -u_2$$

Matrični zapis:

$$\begin{bmatrix} R_1 + R_3 + R_5 & -R_3 \\ -R_3 & R_2 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} u_1 \\ -u_2 \end{bmatrix}$$

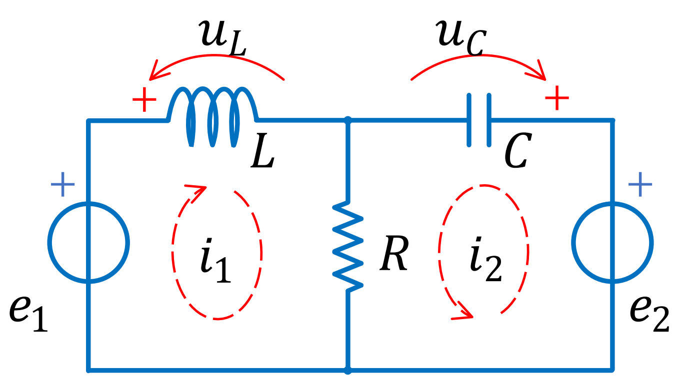
Primer 2 - Električno kolo sa R, L i C

Zadatak:
Napisati model za el.kolo sa slike.



- Formirati matematički model u prostoru stanja za električno kolo na slici.
- Ako se za izlaznu velicinu usvoji struja kroz otpornik R, formirati jednačinu izlaza.
- Formirati jednačinu izlaza ako se za izlazne veličine usvoje struje kroz izvore e_1 i e_2.

Primer 2 – nastavak ...



$$i_2 = C\frac{du_C}{dt}$$
$$u_L = L\frac{di_1}{dt}$$

- Konturne struje

$$e_2 - R(i_1 + i_2) - u_C = 0 \quad \rightarrow \quad R(i_1 + i_2) = e_2 - u_C$$
$$e_1 - u_L - R(i_1 + i_2) = 0 \quad \rightarrow \quad e_1 - u_L - e_2 + u_C = 0$$

- Uvedemo izvode

$$R\left(i_1 + C\frac{du_C}{dt}\right) = e_2 - u_C$$
$$e_1 - L\frac{di_1}{dt} - e_2 + u_C = 0$$

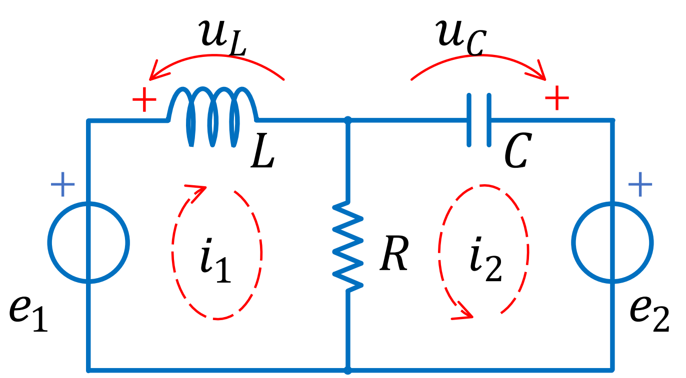
- prepišemo ...

$$\frac{du_C}{dt} = -\frac{1}{RC}u_C - \frac{1}{C}i_1 + \frac{1}{RC}e_2$$
$$\frac{di_1}{dt} = \frac{1}{L}u_C + \frac{1}{L}e_1 - \frac{1}{L}e_2$$

- Matrični zapis

$$\begin{bmatrix} \frac{di_1}{dt} \\ \frac{du_C}{dt} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_1 \\ u_C \end{bmatrix} + \begin{bmatrix} \frac{1}{L} & -\frac{1}{L} \\ 0 & \frac{1}{RC} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

Primer 2 – nastavak 2 ...



$$i_2 = C\frac{du_C}{dt}$$
$$u_L = L\frac{di_1}{dt}$$

- Izlaz: struja kroz otpornik

$$i = i_1 + i_2$$

$$R(i_1 + i_2) = e_2 - u_C \quad \rightarrow \quad i = \frac{e_2 - u_C}{R}$$

$$i = \begin{bmatrix} 0 & -\frac{1}{R} \end{bmatrix} \begin{bmatrix} i_1 \\ u_C \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{R} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

- Izlazi: struje kroz naponske izvore

$$i_2 = C\frac{du_C}{dt} = -\frac{1}{R}u_C - i_1 + \frac{1}{R}e_2$$

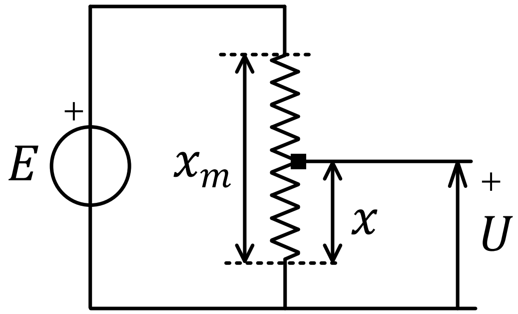
$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & -\frac{1}{R} \end{bmatrix} \begin{bmatrix} i_1 \\ u_C \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{R} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

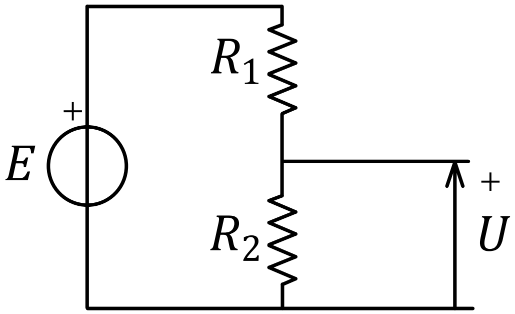
Elektro-mehanički sistemi

Interakcija električnih i mehaničkih sistema

- Električni i mehanički deo sistema mogu imati različite vidove interakcije:
  - mehanička promena otpornosti,
  - kretanje strujnog provodnika u magnetnom polju.

Potenciometar





$$R(x) = \frac{\rho}{A} x$$

$$R(x) = \frac{R_m}{x_m} x$$

$$U(t) = \frac{R_2}{R_1 + R_2} E(t)$$

Promenljive

Promenljive od značaja:

- f_e – elektromagnetna sila $[N]$
- v – brzina provodnika u odnosu na magnetno polje $[m/s]$
- l – dužina provodnika u magnetnom polju $[m]$
- Φ – magnetni fluks $[Wb]$ (izražen u Veberima)
- B – magnetna indukcija $[Wb/m^2]$
- i – jačina električne struje $[A]$
- e_m – indukovana elektromotorna sila $[V]$

Kretanje provodnika u magnetnom polju

- Na pravolinijski provodnik dužine l, kroz koji protiče struja jačine i i koji se kreće u homogenom magnetnom polju indukcije B deluje sila (Amperova sila) intenziteta

$$\vec{f_e} = \overrightarrow{il} \times \vec{B} = ilB \sin\theta$$

θ ugao koji zaklapaju silnice magnetnog polja i smer proticanja struje

Za $\theta = 90°$:

$$f_e = ilB$$

- kretanje provodnika brzinom v u homogenom magnetnom indukuje elektromotornu silu

$$e_m = l\vec{v} \times \vec{B}.$$

Kada se provodnik kreće normalno na silnice magnetnog polja

$$e_m = lvB$$

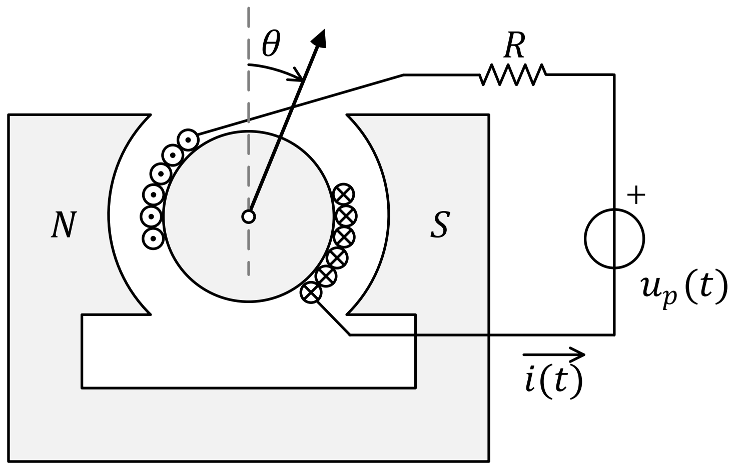
- Dvosmerno dejstvo:

$p_m = f_e v = (Bli)v$ — mehanička snaga - magnetno polje deluje na mehaničke elemente

$p_e = e_m i = (Blv)i$ — električna snaga - proticanje struje usled e_m

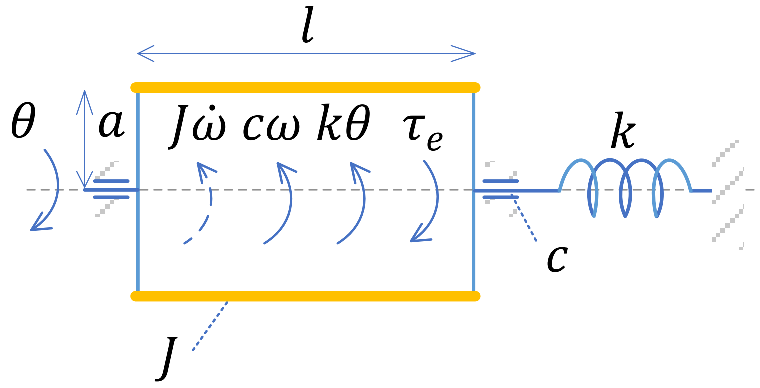
$p_m = p_e$ — iz zakona održanja energija.

Primer 1 - Galvanometar



- Ugaoni pomeraj kalema je srazmeran struji koja protekne kroz namotaj
- U osi kalema postoji torziona opruga (nije nacrtana)
- Pretpostavka: u vazdušnom procepu je uniforman mag. fluks gustine *B*
- Ima n namotaja žice (dužine l, poluprečnika a)

Primer 1 – nastavak ...



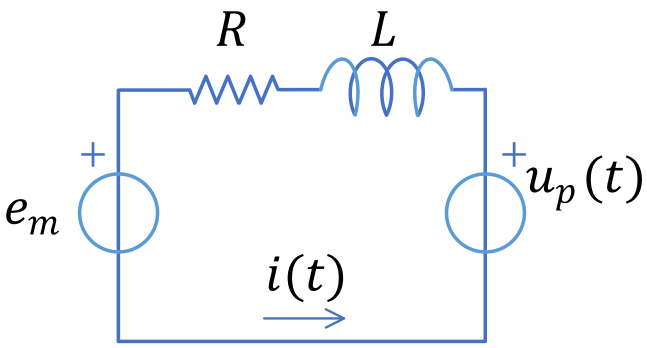
Mehanički deo

$$f_e = (2nl)Bi$$ sila na provodnike u namotaju

$$\tau_e = f_e a = (2nl)Bia$$ moment sile

Ravnoteža momenata sila

$$J\dot{\omega}(t) + c\omega(t) + k\theta(t) = (2nlBa)i(t)$$



Električni deo

$$e_m = (2nl)vB = (2nl)(a\omega)B$$ Indukovana ems

El. kolo

$$L\frac{di(t)}{dt} + Ri(t) + (2nlBa)\omega(t) = u_p(t)$$

Primer 1 - nastavak 2

prepisano …

$$J\dot{\omega}(t) + c\omega(t) + k\theta(t) = (2nlBa)i(t)$$

$$L\frac{di(t)}{dt} + Ri(t) + (2nlBa)\omega(t) = u_p(t)$$

Uvedemo: $\alpha = 2nlBa$

Model (sređeno):

$$\underbrace{\begin{aligned} \frac{d\theta}{dt}(t) &= \omega(t) \\ \frac{d\omega}{dt}(t) &= -\frac{c}{J}\omega(t) - \frac{k}{J}\theta(t) + \frac{\alpha}{J}i(t) \\ \frac{di}{dt}(t) &= -\frac{R}{L}i(t) - \frac{\alpha}{L}\omega(t) + \frac{1}{L}u_p(t) \end{aligned}}_{\text{Model 3. reda}}$$

Ako zanemarimo induktivnost kalema $L \ll 1$

$$\underbrace{L\frac{di(t)}{dt}}_{0} + Ri(t) + \alpha\omega(t) = u_p(t)$$

$$i(t) = -\frac{\alpha}{R}\omega(t) + \frac{1}{R}u_p(t)$$

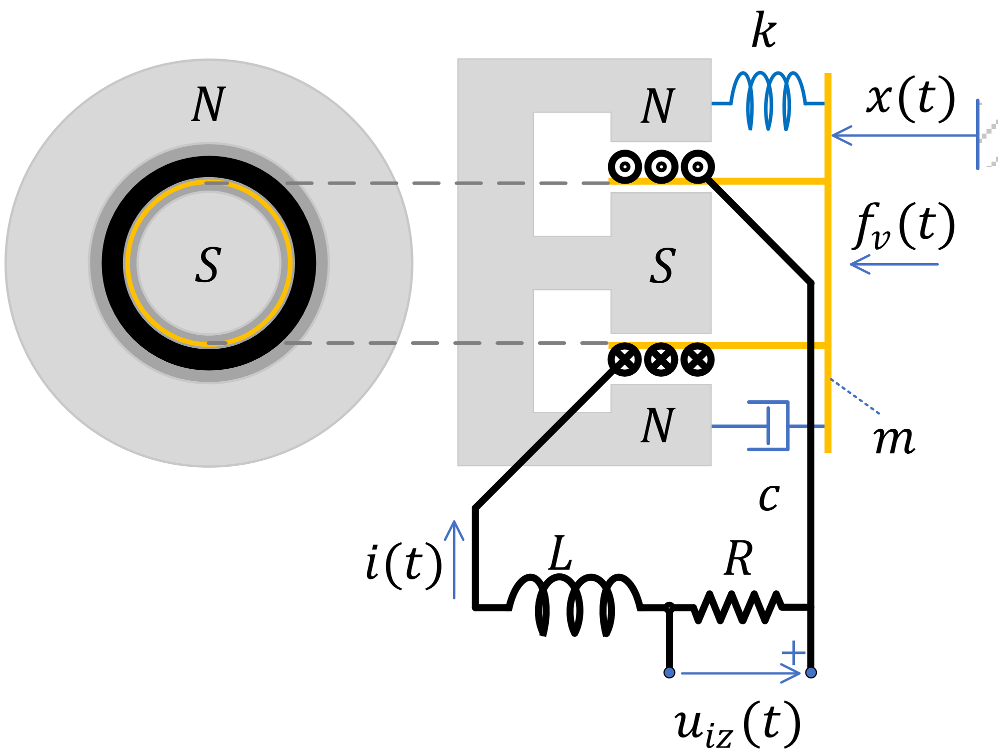
$$\underbrace{\ddot{\theta}(t) + \left(\frac{c}{J} + \frac{\alpha^2}{JR}\right)\dot{\theta}(t) + \frac{k}{J}\theta(t) = \frac{\alpha}{JR}u_p(t)}_{\text{Model 2. reda}}$$

Analiza ponašanja:

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

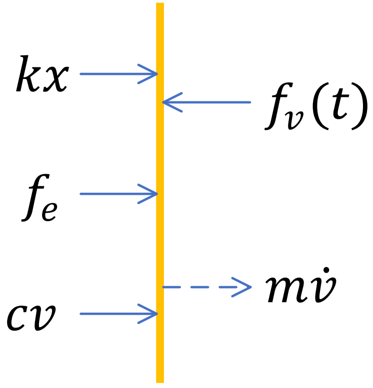
$$\xi = \frac{1}{2\sqrt{kJ}}\left(c + \frac{\alpha^2}{R}\right), \qquad \omega_n = \sqrt{\frac{k}{J}}\ .$$

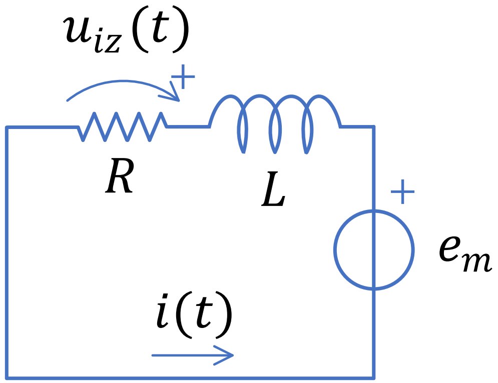
Primer 2 - Mikrofon



- Ima n namotaja žice u kalemu poluprečnika a
- Otpornost namotaja se zanemaruje
- Radijalno mag. polje na namotaje

Primer 2 – nastavak





$$f_e = lBi(t) = (2\pi an)Bi(t) = \alpha \cdot i(t)$$
$$e_m = lBv = (2\pi an)Bv(t) = \alpha \cdot v(t)$$

$$\alpha = (2\pi an)B$$

Mehanički deo: $m\dot{v}(t) + cv(t) + kx(t) = -\alpha i(t) + f_v(t)$

Električni deo: $L\dfrac{di}{dt} + Ri(t) = \alpha v(t)$

Model:

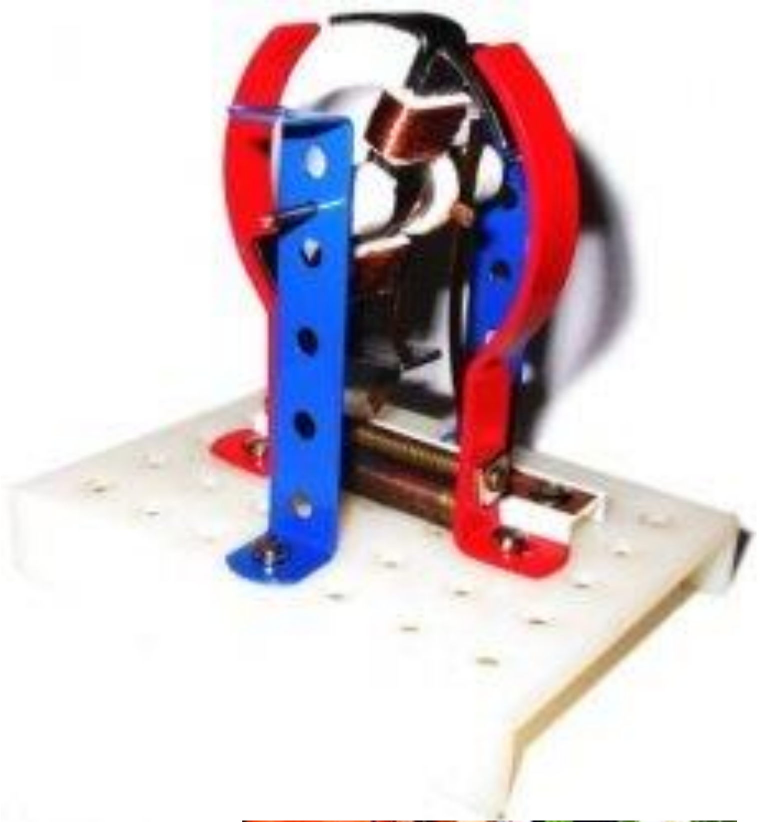
$$\frac{dx(t)}{dt} = v(t)$$
$$\frac{dv(t)}{dt} = \frac{1}{m}\left(-kx(t) - cv(t) - \alpha i(t) + f_v(t)\right)$$
$$\frac{di(t)}{dt} = \frac{1}{L}\left(\alpha v(t) - Ri(t)\right)$$

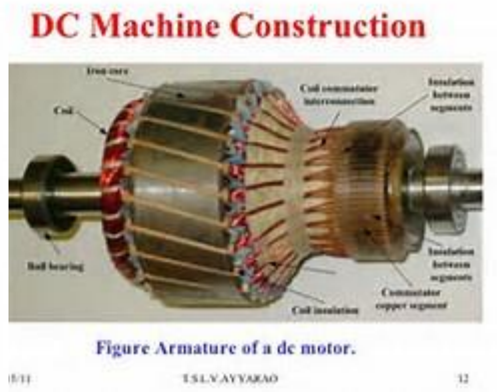
$$u_{iz}(t) = Ri(t)$$

Matrični zapis:

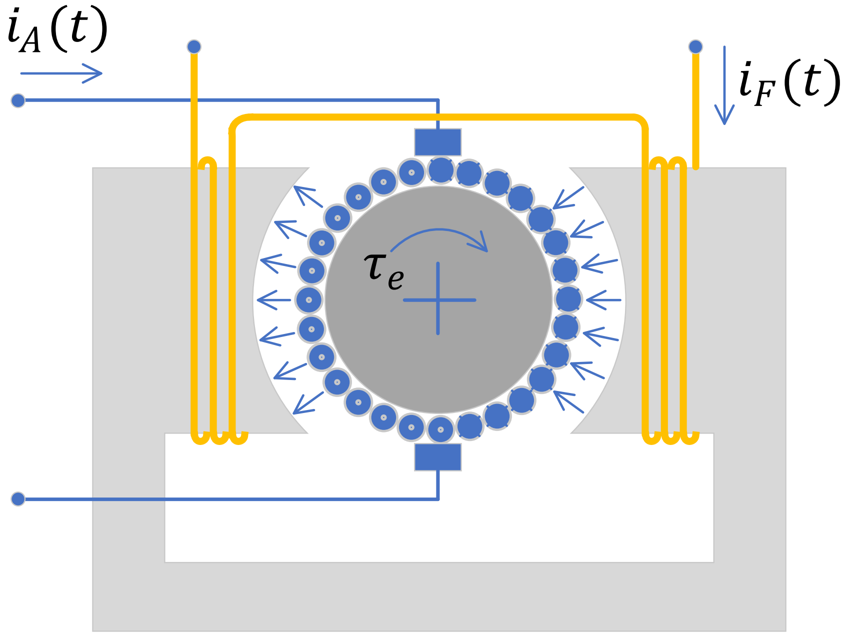
$$\dot{\boldsymbol{x}}(t) = \begin{bmatrix} 0 & 1 & 0 \\ -\frac{k}{m} & -\frac{c}{m} & -\frac{\alpha}{m} \\ 0 & \frac{\alpha}{L} & -\frac{R}{L} \end{bmatrix}\boldsymbol{x}(t) + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{m} \end{bmatrix} f_v(t)$$
$$u_{iz}(t) = \begin{bmatrix} 0 & 0 & R \end{bmatrix}\boldsymbol{x}(t) + 0 \cdot f_v(t)$$

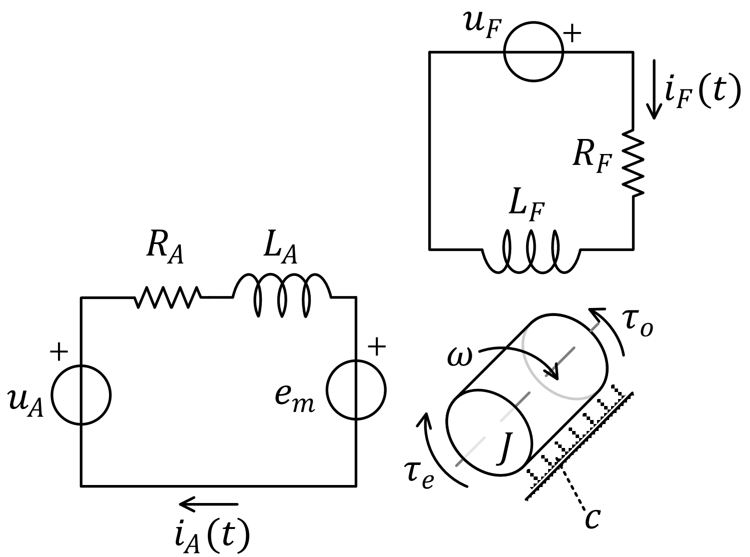
Elektromotor





Primer 3 - Motor jednosmerne struje





- Prikazan je jednosmerni motor sa nezavisnom pobudom, upravljan strujom rotora.
- Odrediti zavisnost brzine rotora od napona rotora i promene opterećenja motora.
- Pretpostaviti da se radi o opsegu brzina do nominalne ($\omega \quad \omega_n$) i da je fluks u mašini $\Phi_F = const$ (pobudna struja $i_F = I_F = const$).

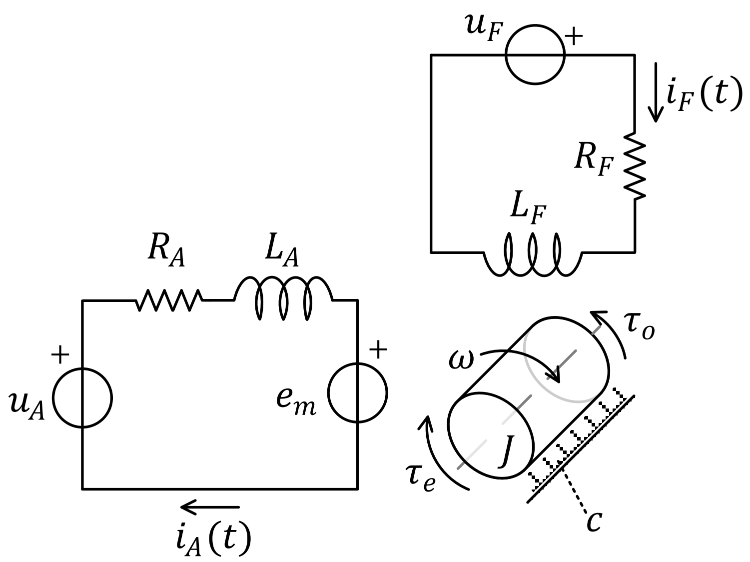
Primer 3 – nastavak ...

$$B = \frac{\Phi(i_F(t))}{A}$$

$$\tau_e(t) = f_e a = \left(l \cdot B \cdot i_A(t)\right)a = \left(l \cdot \frac{\Phi(i_F(t))}{A} \cdot i_A(t)\right)a = \gamma \cdot \Phi\left(i_F(t)\right) \cdot i_A(t)$$

$$\gamma = \frac{la}{A}$$

$$e_m(t) = lvB = l \cdot \omega(t) a \cdot \frac{\Phi\left(i_F(t)\right)}{A} = \gamma \cdot \phi\left(i_F(t)\right) \cdot \omega(t)$$



Ako je: $u_F(t) = U_F = const,\ I_F = \frac{U_F}{R_F},\ \ \alpha = \gamma\Phi(I_F)$

$$\tau_e(t) = \alpha \cdot i_A(t)$$

$$e_m(t) = \alpha \cdot \omega(t)$$

$$\frac{di_A(t)}{dt} = \frac{1}{L_A}\left[-R_A i_A(t) - \alpha\omega(t) + u_A(t)\right]$$

$$\frac{d\omega(t)}{dt} = \frac{1}{J}\left[\alpha i_A(t) - c\omega(t) - \tau_o(t)\right]$$

Primer 3 – nastavak 2

prepisano ...

$$\frac{di_A(t)}{dt} = \frac{1}{L_A}[-R_A i_A(t) - \alpha\omega(t) + u_A(t)]$$

$$\frac{d\omega(t)}{dt} = \frac{1}{J}[\alpha i_A(t) - c\omega(t) - \tau_o(t)]$$

.

U matričnom zapisu:

$$\begin{bmatrix} \frac{di_A(t)}{dt} \\ \frac{d\omega(t)}{dt} \end{bmatrix} = \begin{bmatrix} \frac{-R_A}{L_A} & \frac{-\alpha}{L_A} \\ \frac{\alpha}{J} & \frac{-\mathrm{c}}{J} \end{bmatrix} \cdot \begin{bmatrix} i_A(t) \\ \omega(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L_A} & 0 \\ 0 & -\frac{1}{J} \end{bmatrix} \cdot \begin{bmatrix} u_A(t) \\ \tau_o(t) \end{bmatrix}$$

Model 2. reda == 2 prom. stanja;
2 ulaza

Ako se zamenari induktivnost namotaja $L \ll 1$

$$\frac{d\omega(t)}{dt} = -\left(\frac{\alpha^2}{JR_A} + \frac{\mathrm{c}}{J}\right)\omega(t) + \frac{\alpha}{JR_A}u_A(t) - \frac{1}{J}\tau_o(t)$$

Model 1. reda == 1 prom. stanja;
2 ulaza